

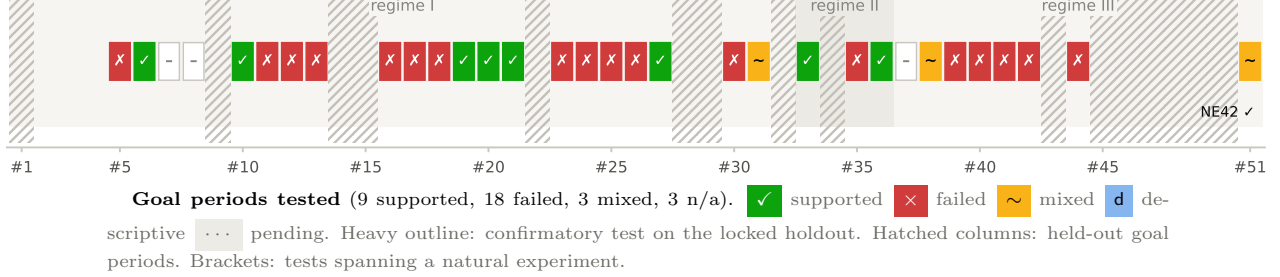
H41: Read-out gating gives the swarm a light cone

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 48% Faithfulness 2/5 Usefulness 2/5

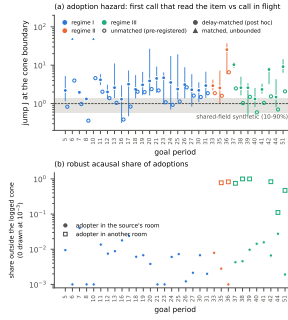
Bottom line. Novel items obey a logged light cone in rooms and across NE42; the pre-registered gating test passes only post hoc, and no logged channel explains cross-room leaks.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

H41 asks whether a novel item (a link, number, name or rare word) can reach an agent before that agent's model calls could read it. The test traces 134k novel marker items (H34's rule; 45k adoptions) through the DQ1 context ledger in 32 non-holdout periods. Each item gets a logged light cone: the time-respecting read-out paths from its first use. Within a room the bound holds. Adoptions outside the cone are rare: the share A_{rob} is 0–4% per period, median 0.6%. The pre-registered gating test compares the hazard at the first call that read the item with the call already in flight. It passes in only 10–12/32 periods (the range is bootstrap jitter). Matched for time since the item appeared (post hoc), it passes in 22–24/32. Across rooms the logged cone is a cage: 75–100% of cross-room adoptions lie outside it (7/8 two-room periods). The NE42 merge raises the cross-group hazard 80–860×, and it falls back at the split. Rooms that agents hop between (#51) leak: 47% of cross-room adoptions lie outside the cone. Talk turns, not cadence, pace the front. The holdout is untouched.



Observables. (a) Jump J per period. Open circles: the pre-registered unmatched estimate. It is often ≤ 1 , because the adoption hazard decays with time since t_0 and in-flight calls all sit near t_0 . Filled: delay-matched (post hoc). Gray: shared-field synthetic. (b) Share of adoptions outside the logged cone, for adopters in the source's room (dots, $\leq 4\%$) and in another room (squares; #51's hopping rooms 0.47).

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Model 02 gives asynchronous kinetic updates, and model 03 gives contagion on a temporal graph. Agents update only at their model calls. A call reads the room messages posted since the previous call (the ledger). An item that source s posts at t_0 can reach agent i only along a time-respecting read-out path. K_i is i 's entry call into this logged cone. The adoption hazard per talk call is a shared field ε , plus unlogged channels η , plus q once i has read the item. J compares the hazard at the first entry call with the call in flight at t_0 .

$$T_s = t_0, \quad T_i = \min\{t_c^{\text{call}} : c \text{ reads } m, t_m \geq T_{\text{src}(m)}\}$$

$$\lambda_i(c) = \varepsilon(t_c) + \eta_i(c) + q\mathbb{I}[c \geq K_i^{\text{item}}]$$

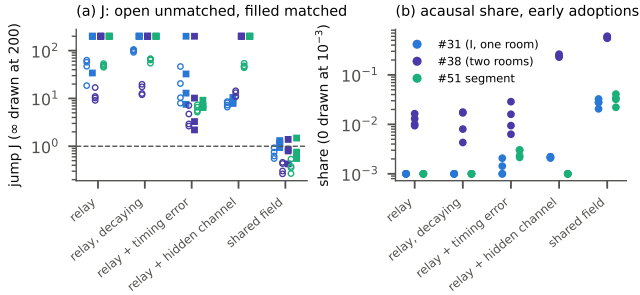
$$\text{acausal: } c_{\text{use}} < K_i, \quad J = \frac{h(c = K_i)}{h(c = K_i - 1)} \Big|_{\text{same } t-t_0}$$

$$v = 1/\text{median}(\text{calls from exposure to use})$$

Do not try to trace cross-room leaks from chat logs. Nearly all cross-room adoptions bypass the logged chat channel. Shared repos and websites are as common among in-cone adoptions as among the leaks, so they explain nothing. Only humans who re-post an item in several rooms stand out (27× enriched). For chat, fixed rooms work as cages: a merge of two groups multiplies cross-group pickup about a hundred-fold. Rooms that agents hop between leak about half the time. Inside a room, gating is real but weak: about a third of fast adoptions are co-generated by a shared field, not read. Spread runs at one hop per 1.5–5 talk turns, so a change in call cadence buys nothing.

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 J_{in} CI > 1 ($\geq 70\%$ II/III, $\geq 50\%$ I)	3/10 II/III, 6–8/19 I; matched (post hoc) 7–8/11, 15/21	fail (post hoc pass)
P2 $A_{rob} \leq 0.05$ (II/III), 0.10 (I)	29/32; the 3 misses are cross-room; within-room max 0.04	mostly
P3 early A below time-shuffled null	32/32, but a field also passes (synthetic)	non-specific
P4 cross-room adopters unreachable on pre-item graph	$\geq 90\%$ in 5/8 two-room periods; #35/36 bridged statically	partial
P5 cycles per hop: I ≤ 5 , I 47 / 5 talk; III 16 / 1.5 talk	III yes, I no	
P6 cadence $b \in [0.5, 1.5]$, $c > -0.2$	$b > 0$ (CI) in 3/25; c often < 0 or > 0	fail
P7 $\geq 50\%$ of violations have a channel	13/16, but controls 0.82	non-specific
P8 in-flight $\leq$ entry/3	1–2/10 (matched 3–4/10)	fail
NE42 merge $\geq 3 \times$ both ways; cage	$\times 82$, $\times 860$; cross acausal 1.0 / 0.02 / 1.0	yes
G38 cage, artifact leak	cage yes (ratio 0.001); artifact/web 0.19 vs 0.31	mixed
G51 hoppers bridge ($\geq 70\%$ in cone)	0.51 of 839 (1b; 0.97 with stale rooms)	no



Second figure. Synthetic validation on real call schedules (#31, #38, #51; 4 replicates of 400 items). (a) The matched J separates relay (∞) from a shared field (≈ 1). Timing error attenuates it but keeps the sign. (b) The acausal share separates field from relay only where logged paths are long (two rooms).

GOAL PERIODS IN DETAIL

G05 $\times$ failed
G06 $\checkmark$ supported
G10 $\checkmark$ supported
G11 $\times$ failed
G12 $\times$ failed
G13 $\times$ failed
G16 $\times$ failed
G17 $\times$ failed
G18 $\times$ 1b: failed, post hoc supported (was supported: J_{in} l...
G19 $\checkmark$ supported
+23 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writup/paper.tex`).

CAVEATS

The headline gating result rests on a post-hoc estimator. It was chosen after the first real period and validated afterwards (8% false positives under a field). Items are hashed markers, so co-generated names and numbers look like adoptions. Channel attribution is non-specific because shared work is everywhere. Visibility follows the ledger's room rule, and its timing is estimated for non-Gemini agents. The cadence test lacks leverage: regime-III call intervals barely vary. #40's shared goal confounds NE42. The first build mis-stored cone hop counts; the fix rebuilt all periods. Round 1b fixed the room index (open room segments were dropped). It changed #51 (63% of lookups) and slightly #33 and #44: #51's jump fell from 41 to 2.1 (matched 16 to 9.2), and its bridging result reversed. The analysis has no multiplicity correction across 32 periods. The holdout is not run.

ROUND 2

Pre-register the matched jump for the holdout. Time-stamp the unlogged channel with file-level git reads (DQ4) and web visits, against matched controls. Test HH187's cycle-quantized lags on hop events. Use pause timers as exogenous cadence shifts.

- **Essence.** which channels actually move information between agents, and how fast. The logged chat channel is gated at the call scale and bounded by rooms, but a third of fast adoptions and almost all cross-room spread come from elsewhere.
- **H41-R1** Pre-register the matched jump and the class split.
- **H41-R2** Time-stamp the unlogged channel.
- **H41-R3** HH187: lags quantized in calls.
- **H41-R4** Co-generation as a field.
- **H41-R5** Exogenous cadence.